

1) Given : $\dot{m} = 15 \text{ kg/s}$

$$P_1 = 12 \text{ MPa}$$

$$T_1 = 640 \text{ K}$$

$$P_2 = 1 \text{ MPa}$$

$$x_2 = 0.86$$

Assume : S.L.S. $+1/2$

: Steady-flow $+1/2$

: quasi-ef. $+1/2$

: $\Delta h_e = \Delta h_c = \dot{Q} = 0$ $+1/2$

a) Steady flow : $\dot{m}(\Delta h + \Delta h_e + \Delta h_c) = \dot{Q} - \dot{W}$ $+2$

Turbine : $\dot{m}(h_2 - h_1) = -\dot{W}$

$\dot{W} = -\dot{m}(h_2 - h_1) \rightarrow$ State 1 : $P_1 = 12 \text{ MPa}$

$$T_1 = 640 \text{ K}$$

Table D.2 : $T_1 > T_{\text{sat}}$ - Vapor

Table D.3Q : $h_1 = 2926 \frac{\text{kJ}}{\text{kg}}$ $+1$

$$s_1 = 5.8843 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$\rightarrow$ State 2 : $P_2 = 1 \text{ MPa}$

$$x_2 = 0.86$$

$$h_2 = h_g x_2 + (1 - x_2) h_f$$
 $+1$

$$= (0.86)(2777.1) + (1 - 0.86)(762.52)$$

$$= 2495.1 \frac{\text{kJ}}{\text{kg}}$$
 $+1$

$$s_2 = s_g x_2 + (1 - x_2) s_f$$
 $+1$

$$= (0.86)(6.585) + (1 - 0.86)(2.1381)$$

$$= 5.963 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$
 $+1$

$$\rightarrow \dot{W} = -(15)(2495.1 - 2926) \rightarrow \boxed{\dot{W} = 6463.5 \text{ kW}} \quad +1$$

$$\Delta S = 5.963 - 5.8843 \rightarrow \boxed{\Delta S = 0.0787 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}} \quad +1$$

b) ideal turbine: $S_2 = S_1 = 5.8843 \text{ kJ/kg} \cdot \text{K} +2$

$$x_{2,s} = \frac{5.8843 - 2.1381}{6.585 - 2.1381} \rightarrow x_{2,s} = 0.842 +1$$

$$\begin{aligned} h_{2,s} &= x_{2,s} h_g + (1 - x_{2,s}) h_f \\ &= (0.842)(2777.1) + (1 - 0.842)(762.52) \\ &= 2458.796 \text{ kJ/kg} +1 \end{aligned}$$

$$\dot{W}_{\text{ideal}} = -\dot{m}(h_{2,s} - h_1) = -15(2458.796 - 2926)$$

$$\boxed{\dot{W}_{\text{ideal}} = 7008.06 \text{ kW}} +1$$

$$\eta_{\text{turb}} = \frac{\dot{W}_{\text{real}}}{\dot{W}_{\text{ideal}}} = \frac{6463.5}{7008.06} \rightarrow \boxed{\eta_{\text{turb}} = 0.92} +1$$

c) Turbine is possible if $S_2 \geq S_1$ +2

@ $x_2 = 0.8$, $S_2 = 5.696 \text{ kJ/kg} \cdot \text{K}$, which is less than S_1

$\Rightarrow$ cannot have a turbine where $x_2 = 0.8$ for this initial condition and final pressure +3

[2] Given: P_1 [psia]

P_2 [psig]

T_1, T_2

Assume: ideal gas, $c_p = \text{const}$, steady flow, quasi- $\Delta ke = \Delta pe = \dot{Q} = 0$ +2 for assume

$$P_2 \text{ [psia]} \rightarrow P_2 \text{ [psia]} = P_2 \text{ [psig]} + P_1 \text{ [psia]} +2$$

$$\Rightarrow \eta_{\text{comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} +2$$

$\rightarrow$ because steady flow: $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

because compressor: $\dot{m}(h_2 - h_1) = -\dot{W}$

because ideal gas $c_p = \text{const}$: $\dot{m}c_p(T_2 - T_1) = -\dot{W} +2$

$\rightarrow \dot{W}_{\text{ideal}} = -\dot{m}c_p(T_{2,s} - T_1)$ where $T_{2,s} = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} +2$

$$\dot{W}_{\text{real}} = -\dot{m}c_p(T_2 - T_1) \Rightarrow \boxed{\eta_{\text{comp}} = \frac{T_{2,s} - T_1}{T_2 - T_1}} +3 \text{ for plot} +2 \text{ for discussion}$$

[3] Heat pump $\rightarrow$ goal is to move heat to a warm space +3
 $\Rightarrow \beta = \frac{\dot{Q}_H}{\dot{W}_{in}}$ +2

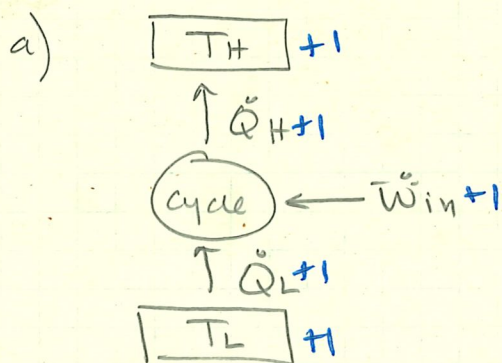
Refrigerator $\rightarrow$ goal is to remove heat from a cold space +3
 $\Rightarrow \beta = \frac{\dot{Q}_L}{\dot{W}_{in}}$ +2

[4] Given: $\dot{W}_{in} = 1000 \text{ W}$

Assume: steady-state

$$T_L = 0^\circ\text{C} = 273.15 \text{ K} \quad +1$$

$$T_H = 20^\circ\text{C} = 293.15 \text{ K} \quad +1$$



b) $\beta_{max} = \frac{T_L}{T_H - T_L}$ +2

$$= \frac{273.15}{20}$$

$$\beta_{max} = 13.658 \quad +1$$

c) $\beta_{real} = 7$ where $\beta = \frac{\dot{Q}_L}{\dot{W}_{in}}$ +2

$$\rightarrow \dot{Q}_L = \beta \dot{W}_{in} = (7)(1000) \rightarrow \dot{Q}_L = 7000 \text{ W} \quad +1$$

$$\rightarrow \dot{Q}_H = \dot{Q}_L + \dot{W}_{in} = 1000 + 7000 \rightarrow \dot{Q}_H = 8000 \text{ W} \quad +1$$

[5] Given: $x_3 = 1$

$P_3 = 8 \text{ MPa}$

$\dot{W}_{\text{net}} = 100 \text{ MW}$

$P_4 = 0.008 \text{ MPa} = P_1$

$x_1 = 0$

$\eta_{\text{pump}} = \eta_{\text{turb}} = 0.85$

Assume: S.C.S.

: quasi eq

: steady-flow

→ pump/turbine: $\Delta K.E. = \Delta P.E. = \dot{Q} = 0$

→ boiler/condenser: $\Delta K.E. + \Delta P.E. = \dot{W} = 0$

+3 for assume

⇒ find: $\dot{m}$, $\dot{W}_{\text{turb}}$

→ Cycle: 1-2 → compression: $\dot{m}(h_2 - h_1) = -\dot{W}_{\text{pump}} + 2$

2-3 → boiler: $\dot{m}(h_3 - h_2) = \dot{Q}_{\text{boiler}} + 1$

3-4 → expansion: $\dot{m}(h_4 - h_3) = -\dot{W}_{\text{turb}} + 2$

4-1 → condenser: $\dot{m}(h_1 - h_4) = \dot{Q}_{\text{cond}} + 1$

→ States

	<u>P</u>	<u>Phase</u>	
1	0.008 MPa	$x_1 = 0$	] $\eta_{\text{pump}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} = \frac{h_{2,s} - h_1}{h_2 - h_1} + 1$
2	8 MPa	liquid	
3	8 MPa	$x_3 = 1$	] $\eta_{\text{turb}} = \frac{\dot{W}_{\text{real}}}{\dot{W}_{\text{ideal}}} = \frac{h_4 - h_3}{h_{4,s} - h_3} + 1$
4	0.008 MPa	mixture	

⇒ $\dot{W}_{\text{net}} = \dot{W}_{\text{pump}} + \dot{W}_{\text{turb}} = 100 \text{ MW} + 1$

$= -\dot{m}(h_2 - h_1) + \dot{m}(h_4 - h_3)$

⇒ $\dot{m} = \frac{-\dot{W}_{\text{net}}}{(h_2 - h_1) + (h_4 - h_3)} + 2 \Rightarrow \text{find } h\text{'s to solve for } \dot{m}$

State 1 : $P_1 = 0.008 \text{ MPa}$

$x_1 = 0$

Table D2 $\rightarrow h_1 = h_f = \underline{173.84} \text{ kJ/kg}^{+1}$
 $s_1 = s_f = 0.59249 \text{ kJ/kg}\cdot\text{K}$

State 2s : $P_2 = 8 \text{ MPa}$

$s_{2s} = s_1 = 0.59249 \text{ kJ/kg}\cdot\text{K}^{+1}$

$s_{2s} < s_g \rightarrow \text{liquid}^{+1}$

NIST $\rightarrow h_{2s} = 182.44 \text{ kJ/kg}$ @ $T_{2s} = 315.03 \text{ K}$

State 2 : $\eta_{\text{pump}} = 0.85 = \frac{h_{2s} - h_1}{h_2 - h_1}^{+1}$

$h_2 = h_1 + \frac{h_{2s} - h_1}{0.85} = 173.84 + \frac{(182.44 - 173.84)}{0.85}$

$h_2 = \underline{183.96} \frac{\text{kJ}}{\text{kg}}^{+1}$

State 3 : $P_3 = 8 \text{ MPa}$

$x_3 = 1$

Table D2 $\rightarrow h_3 = h_g = \underline{2758.7} \text{ kJ/kg}^{+1}$
 $\rightarrow s_3 = s_g = 5.7450 \text{ kJ/kg}\cdot\text{K}$

State 4s : $P_4 = 0.008 \text{ MPa}$

$s_{4s} = s_3 = 5.7450 \text{ kJ/kg}\cdot\text{K}^{+1}$

$\Rightarrow s_f < s_{4s} < s_g \rightarrow \text{mixture}$

$x_{4s} = \frac{s_{4s} - s_f}{s_g - s_f} = \frac{5.7450 - 0.59249}{8.2273 - 0.59249} = 0.675^{+1}$

$h_{4s} = x_{4s} h_g + (1 - x_{4s}) h_f$

$= (0.675)(2576.2) + (1 - 0.675)(173.84)$

$h_{4s} = 1795.43 \text{ kJ/kg}^{+1}$

State 4 : $\eta_{\text{turb}} = 0.85 = \frac{h_1 - h_3}{h_{4s} - h_3}^{+1}$

$h_4 = h_3 + 0.85(h_{4s} - h_3)$

$= 2758.7 + 0.85(1795.43 - 2758.7) \rightarrow h_4 = \underline{1939.16} \frac{\text{kJ}}{\text{kg}}^{+1}$

$$\Rightarrow \dot{m} = \frac{-100 \times 10^3}{(183.96 - 173.84) + (1939.6 - 2758.7)} \text{ kW}$$

$$\boxed{\dot{m} = 123.6 \text{ kg/s}} \quad +1$$

$$\begin{aligned} \Rightarrow \dot{W}_{\text{turb}} &= -\dot{m} (h_4 - h_3) \\ &= -123.6 (1939.6 - 2758.7) \end{aligned}$$

$$\boxed{\dot{W}_{\text{turb}} = 101.24 \text{ MW}} \quad +1$$